The Multiple Regression Model Transcript

Hello everyone, welcome to lesson 1 video 1. In this session, we will introduce the concept of multiple linear regression. This model extends simple linear regression into two or more predictor variables by explaining the variance on an outcome variable. We have spent a lot of time in learning how to model or how do we model a relationship between two variables with a simple linear regression.

But the real world or actually in the real world, things are rarely that simple, right? Usually, an outcome is influenced by many different factors working together. Today, we are going to dig a little more deeper into various different factors that can affect one variable and hence we are going to study about multiple linear regression. So, we are going beyond one variable or the effect of one variable on the other variable.

We are going to study how does multiple variables affect one variable. So, we kind of are going to study several predictors and their effects on one variable. Now, to give you an understanding using a simple example, the prices of houses could very well be dependent on the size in terms of the square feet, the number of bedrooms it has, a 2BHK house would be slightly lesser in price than a 3BHK or 4BHK house, right? Similarly, a 1000 square feet home would be slightly lesser in price than a 4000 square feet house, right? So, both of these would affect the prices.

Similarly, the location as well, the location rating or where that house is would also be very very important. Let's say a house in Delhi or a house in Bangalore would be very much affected by the location versus a house which is in tier 2 cities in India, right? So, in total, the price itself in this particular case could be dependent not only on one variable or let's say only on size but it could be dependent on multiple variables like size, bedrooms, location, XYZ and so on. There could be hundreds of these factors.

So, the idea is to understand how this more than one predictor variable or more than one independent variable affects one variable which in this case, in this example is sales price of house, right? This is how we represent the entire equation Y is equals to beta 0 plus beta 1 X1 plus beta 2 X2. This is the equation of a multidimensional line or what we call as hyperplane, right? That is not a line but a line is a hyperplane in many dimensions. So, if we would have been discussing about linear regression, we were only discussing about Y is equals to beta 0 plus beta 1 X1.

Now, because we are extending the relationship of Y not only with X1 but with many other variables, that is why we are generalising the entire line and that line becomes a hyperplane in higher dimension because we are dealing with not only one dimension which is X1, X2 or not only one but multiple dimensions X1, X2, X3 and Xk and so on, right? So, this is what it is. Now, these beta 0, beta 0 is still the intercept term which was there in the simple linear regression and beta 1 to beta k are the coefficient of each and every variable that we introduce.

Now, you can think of beta 1 in this case as the rate of change of Y with respect to X1, beta 2 in this case would be the rate of change of Y with respect to X2 and so on.

So, that is what these are individual slope parameter for each predictor variable, right? So, the most important concept that we have got is you have price which is affected because of the variables size and bedrooms in this case and then we associate the parameters with them, these coefficients. The idea is to get a data and to estimate these coefficients beta 0, beta 1 and beta 2, okay? Now, beta 1 represents the estimated change in price of one unit increase in size. So, if our size goes from 1000 square feet to one more square feet, what does the change in price looks like? Similarly, how does the change in price looks like when the number of bedrooms increases from 1 to 2 to 2 to 3 and so on, right? So, that is what we are going to do.

And that is what we are going to learn in this entire session on multiple linear regression, okay? The relationship between, I told you that the relationship between Y and X variable, it actually would look like hyperplane and this is what the simulation helps you understand. In three dimensions, no more would be a line, right? It would not be a line anymore. It would be a plane.

So, this is that plane if you think of. Similarly, on this website which will be given to you or which would be accessible to you, here if you see, these are your data points in three dimensions. You can think of one dimension is number of bedrooms, the other dimensions of the square feet and the Y axis or one end dimension is the price of the houses.

Now, when you vary with the slope, how does the entire hyperplane changes, right? How does the sum of squared of errors changes? The idea is to minimise that value, right? So, and using in going that direction was maximising. So, we are moving in opposite direction. So, you see that it goes up to, I think, S here, right? Now, you can minimise the slope for the other variable.

So, here it is increasing. We go in the other direction, okay? You also can minimise the intercept term. Now, you see that all of these vertical lines which are nothing but measuring the errors because everything on the plane are the predicted values.

And these red data points are the actual data points. Now, our idea is to make these vertical lines as small as possible. So, we can animate that fit as well.

So, you see here, this is the best possible curve that can fit in this data, okay? You can also, I mean, if you feel you can click on shift and then you can also look into from various different perspective, okay? I hope this is clear that in multiple linear regression, we are fitting a hyperplane, a higher dimensional line in our higher dimensional data point. In summary, multiple linear regression allows us to capture the combined inference of several variables on one variable. It is one of the most widely used tools in applied statistics from economics to health sciences to whatever, right?

In the next video, we will study how to estimate the coefficients of multiple linear regression model using least square methods.

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